

Assignment on Line Coding Techniques

Course: EEE-301 – Data and Telecommunication

Chapter: Digital Transmission (Chapter-4)

Topic: Bit to Digital Signal Conversion (Line Coding)

Problem Statement

Determine the digital signal for the following binary data using:

- Unipolar (NRZ)
- RZ (Return to Zero)
- NRZ-L
- NRZ-I
- Manchester
- Differential Manchester

Q1. Determine digital signal for the following data (i) 1011001001, and (ii) 11011110011 using Unipolar, RZ, NRZ-L, NRZ-I, Manchester, Differential Manchester,

Given data:

(i) 1011001001

(ii) 11011110011

Basic Assumptions (As per PDF)

- Positive voltage = +V
- Negative voltage = -V
- Zero voltage = 0
- Initial signal level (for NRZ-I & Differential Manchester) = **Low (-V)**

1. Unipolar NRZ Encoding

Definition (from PDF):

In Unipolar encoding, only one voltage level is used.

Binary **1** is represented by **+V** and binary **0** by **0 voltage**.

(i) **1011001001**

Bit	1	0	1	1	0	0	1	0	0	1
Signal	+V	0	+V	+V	0	0	+V	0	0	+V

(ii) 11011110011

Bit	1	1	0	1	1	1	1	0	0	1	1
Signal	+V	+V	0	+V	+V	+V	+V	0	0	+V	+V

2. RZ (Return to Zero) Encoding

Definition:

Each bit period is divided into two halves.

- 1 → +V then 0
- 0 → -V then 0

(i) 1011001001

Bit	1	0	1	1	0	0	1	0	0	1
Signal	+V→0	-V→0	+V→0	+V→0	-V→0	-V→0	+V→0	-V→0	-V→0	+V→0

(ii) 11011110011

Signal Pattern
+V→0, +V→0, -V→0, +V→0, +V→0, +V→0, +V→0, -V→0, -V→0, +V→0, +V→0

3. NRZ-L (Non-Return-to-Zero Level)

Definition:

- Binary 1 → -V
- Binary 0 → +V

(i) 1011001001

Bit	1	0	1	1	0	0	1	0	0	1
Signal	-V	+V	-V	-V	+V	+V	-V	+V	+V	-V

(ii) 11011110011

Signal
-V, -V, +V, -V, -V, -V, -V, +V, +V, -V, -V

4. NRZ-I (Non-Return-to-Zero Invert)

Definition:

- 1 → Invert signal
- 0 → No change

(i) 1011001001

(Initial level = -V)

Bit	1	0	1	1	0	0	1	0	0	1
Level	+V	+V	-V	+V	+V	+V	-V	-V	-V	+V

(ii) 11011110011

Level
+V, -V, -V, +V, -V, +V, -V, -V, -V, +V, -V

5. Manchester Encoding

Definition:

There is always a transition in the middle of the bit interval.

- 1 → Low to High
- 0 → High to Low

(i) 1011001001

Bit	1	0	1	1	0	0	1	0	0	1
Signal	LH	HL	LH	LH	HL	HL	LH	HL	HL	LH

(ii) 11011110011

Signal
LH, LH, HL, LH, LH, LH, LH, HL, HL, LH, LH

6. Differential Manchester Encoding

Definition:

- Always a transition at the middle
- 0 → Transition at the beginning
- 1 → No transition at the beginning

(i) 1011001001

(Initial level = High)

Bit	1	0	1	1	0	0	1	0	0	1
Rule	NT	T	NT	NT	T	T	NT	T	T	NT

(ii) 11011110011

Rule
NT, NT, T, NT, NT, NT, NT, T, T, NT, NT

Conclusion

Different line coding techniques represent binary data using different voltage patterns.

- **Unipolar** is simple but suffers from DC component
 - **RZ** improves synchronization
 - **NRZ-L and NRZ-I** are bandwidth efficient
 - **Manchester and Differential Manchester** provide self-synchronization and are widely used in data communication systems
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